

TITAN BOS: COMPLETE TECHNICAL ARCHITECTURE

PWA Chatbots + SPA Control Panels + Full-Screen Apps

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Critical corrections: Proper app types, deployment models, technical stack

I. THE COMPLETE APP PORTFOLIO

TITAN HELLO (Full-Screen App / SPA Control Panel)

Technology: React SPA (Single Page Application)

Deployment: Browser-based (responsive for tablet/desktop)

Audience: Lead gen specialists / marketing agents

What it does:

- └ Web scraping interface (target selection)
- └ Outreach templates (vertical-specific)
- └ Campaign management (drag-drop workflows)
- └ Follow-up sequences (automated)
- └ Lead tracking (conversion metrics)
- └ Booking integration (accepts leads into system)
- └ Full-screen, data-entry focused

Features:

- └ Click cards to run AI prompts
- └ Example: Click "Overdue Follow-ups" card → AI generates personalized emails
- └ Example: Click "Campaign Performance" card → AI analyzes conversion rates
- └ Example: Click "Lead Quality" card → AI suggests targeting improvements
- └ All AI-assisted, user editable

Price: \$299-599/month (vertical-specific)

GROUND ZERO (Full-Screen SPA Control Panel)

Technology: React SPA

Deployment: Browser-based (tablet/desktop optimized)

Audience: Dispatch managers / schedulers

What it does:

- └ Calendar view (drag-drop jobs)
- └ Route optimization visualization
- └ Technician assignment (skill-based)
- └ Real-time dispatch (push to TITAN GO)
- └ Status tracking (live job updates)
- └ Multi-site management (switch between sites)
- └ Full-screen, real-time operations

UI Features:

- └ Large widgets/cards for each function
- └ Click "Unassigned Jobs" card → AI suggests best technician
- └ Click "Route Optimization" card → AI optimizes order
- └ Click "Technician Utilization" card → AI warns of overload
- └ Click "High-Risk Jobs" card → AI flags compliance issues
- └ All AI-assisted, dispatcher can override

Price: \$199-799/month (vertical-specific)

TITAN GO (PWA - Chatbot for Field Staff)

Technology: React Native PWA (Progressive Web App)

Deployment: Installable on phone (app-like experience, no app store needed)

Audience: Field technicians / cleaning teams

What it does:

- └ Conversational interface (chatbot style)
- └ Today's jobs (pushed from GROUND ZERO)
- └ Navigate to next job (GPS navigation)
- └ Follow checklists (conversational)
- └ Upload proof photos (context-aware prompts)
- └ Completion signature
- └ Offline execution (syncs when online)
- └ Mobile-first, works in field

Chatbot Interaction:

- └ "What's my first job?" → Shows job details
- └ "Ready to clean?" → Shows checklist as conversation
- └ "Upload before photos" → Launches camera, stores locally
- └ "Is this done right?" → AI analyzes photo quality

- └ "Sign off" → Signature capture
- └ All via conversational interface (not forms)

Price: \$99-399/month (vertical-specific)

ZERO FUSS (PWA - Customer Chatbot Portal)

Technology: React Native PWA

Deployment: Installable on phone / accessible on web

Audience: Customers (property managers, agents, homeowners)

What it does:

- └ Job status tracking (conversational)
- └ Photo viewing (before/after galleries)
- └ Invoice access (payment options)
- └ Rescheduling requests (conversational)
- └ Issue reporting (auto-escalation)
- └ Upsell recommendations (AI-driven)
- └ Review submission
- └ 100% chatbot interface (zero forms)

Chatbot Interaction:

- └ "Where's my cleaner?" → Shows real-time location + ETA
- └ "Show before/after" → Displays gallery
- └ "How much do I owe?" → Shows invoice with payment link
- └ "I need to reschedule" → AI finds available times
- └ "Something's wrong" → Auto-escalates to staff
- └ "What else can you do?" → AI recommends add-ons
- └ All AI-powered, conversational

Price: \$99-499/month

TITAN PRO (Full-Screen Dashboard App)

Technology: React SPA with D3/Recharts visualizations

Deployment: Browser-based (desktop/large tablet optimized)

Audience: Leadership / business owners

What it does:

- └ KPI dashboards (real-time metrics)
- └ Vertical-specific modules (ManagedPremises, Assets, etc)
- └ Intervention radar (AI alerts)

- └ Multi-site overview
- └ Team performance analysis
- └ Financial reporting
- └ Compliance tracking
- └ Full-screen, analytics-focused

Widget/Card Interactions:

- └ Click "Revenue" card → AI trends analysis
- └ Click "Completion Rate" card → AI identifies issues
- └ Click "Team Utilization" card → AI predicts bottlenecks
- └ Click "Customer Satisfaction" card → AI flags unhappy customers
- └ Click "Overdue Invoices" card → AI suggests collection strategy
- └ Click "Compliance" card → AI verifies regulatory adherence
- └ All AI-assisted, drill-down capability

Price: \$199-799/month (vertical-specific modules)

TITAN ZERO (PWA - Vertical-Specific AI)

Technology: React Native PWA + LLM integration

Deployment: Installable on phone / web

Audience: Team members needing vertical training

What it does:

- └ Vertical-specific AI training (context-aware)
- └ Document generation (bond packs, incident reports, SWMS)
- └ Procedural memory (learns your business)
- └ Optimization recommendations
- └ Compliance guidance
- └ Best practices per vertical
- └ Conversational interface

Chatbot Interaction:

- └ "How do I clean a bond property?" → AI explains step-by-step
- └ "What should I check?" → AI lists inspection points
- └ "Generate bond pack" → AI creates document with photos
- └ "I found damage" → AI walks through documentation
- └ "Is this compliant?" → AI checks against regulations
- └ All vertical-specific, learned from training data

Price: \$149-399/month (vertical-specific)

ZEROPAY (PWA - Zero-Fee Payments + AI Follow-up)

Technology: React Native PWA + payment integration

Deployment: Installable on phone / web

Audience: Customers paying + admin managing collections

What it does:

- └ Payment link sending (SMS, email, portal)
- └ Multiple payment methods (PayID, bank, card)
- └ Zero transaction fees (company side)
- └ AI-powered late payment follow-up
- └ Payment attempt tracking
- └ Reconciliation assistance
- └ Mobile/web accessible

AI Follow-up Automation:

- └ Day 3: AI generates SMS reminder (personalized)
- └ Day 7: AI generates email (different angle)
- └ Day 14: AI suggests phone call script
- └ Day 21: AI proposes payment plan
- └ Day 30: AI escalation recommendation
- └ All AI-written, human-approvable

Customer Payment Experience:

- └ Click payment link
- └ Choose method (PayID = zero fees, card = customer pays)
- └ Instant confirmation
- └ No company transaction fees ever

Price: FREE (with option for PRO \$49 with AI follow-up)

TITAN STUDIO (Full-Screen App for Agents)

Technology: React SPA with template engine + media editor

Deployment: Browser-based (desktop optimized)

Audience: Marketing agents / content creators

What it does:

- └ Social media content creation (AI-assisted)
- └ Landing page builder (drag-drop, AI-powered)
- └ Email campaign templates
- └ Newsletter scheduling
- └ Review management
- └ Referral program setup

- └ Analytics dashboard
- └ Full-screen, creative-focused

AI-Assisted Workflow:

- └ Click "Create Social Post" → AI writes 3 options, you edit
- └ Click "Build Landing Page" → AI auto-populates from company data
- └ Click "Email Campaign" → AI suggests timing, subject line
- └ Click "Schedule Newsletter" → AI recommends content
- └ Click "Manage Reviews" → AI suggests responses
- └ Click "Run Campaign" → AI tracks performance

All widgets are clickable to trigger AI:

- └ Every card/widget has "Powered by AI" option
- └ Edit after AI generates
- └ Publish when ready
- └ Track performance via dashboards

Price: \$249/month (shared across verticals)

TITAN SOLO (PWA - Solo Operator Mode)

Technology: React Native PWA (simplified)

Deployment: Installable on phone (primary) / web

Audience: Solo technicians / independent operators

What it does:

- └ Today's jobs (map view primary)
- └ Quick job lookup
- └ Simplified checklists
- └ Fast photo capture
- └ Quick invoice creation
- └ Simple scheduling view
- └ Income tracking
- └ Mobile-first, minimal screens

Chatbot-Style Interface:

- └ "What's next?" → Shows next job on map
- └ "Navigate" → GPS to job
- └ "Ready?" → Starts checklist conversation
- └ "Take photos" → Streamlined capture
- └ "Done" → Quick invoice generation
- └ "How much today?" → Income tracker
- └ No team management (solo focus)

No multi-screen complexity:

- └ Routes through map
- └ Conversations not forms
- └ Minimal typing
- └ Fast completion

Price: \$29-49/month

II. DEPLOYMENT SUMMARY

PWA Apps (Installable on Phone, Work Offline)

TITAN GO	→ Chatbot for field staff (offline-first)
ZERO FUSS	→ Customer portal (chatbot)
TITAN ZERO	→ Vertical AI training (chatbot)
ZEROPAY	→ Payment + follow-up (PWA)
TITAN SOLO	→ Solo tech mode (simplified PWA)

SPA Control Panels (Full-Screen Desktop/Tablet)

TITAN HELLO	→ Lead gen management (full-screen SPA)
GROUND ZERO	→ Dispatch control (full-screen SPA, real-time)
TITAN PRO	→ Admin dashboard (full-screen SPA, analytics)
TITAN STUDIO	→ Marketing automation (full-screen SPA, editor)

Technology Stack

Frontend:

PWA Apps:	React Native Web + Progressive Web App
SPA Control Panels:	React + TypeScript + Redux
UI Components:	Tailwind CSS + shadcn/ui
Maps/Charts:	Google Maps API + Recharts/D3
Mobile:	Progressive Web App (no app store)
Real-time:	WebSocket for live updates

Backend:

Framework:	Laravel (MagicAI v10)
Database:	MySQL + SQLite (for offline sync)
Authentication:	JWT (same across all apps)

API:	REST + GraphQL
Real-time:	WebSocket / Server-Sent Events
File Storage:	S3 (photos, documents)
AI Integration:	Vertex AI, Claude, etc.

Infrastructure:

Hosting:	Cloud (AWS/GCP)
CDN:	CloudFlare
Mobile:	Installable via PWA (no app store)
Offline:	Service Workers + Local Storage
Sync:	Automatic when online

III. HOW USERS EXPERIENCE IT

SOLO CLEANER (Day 1)

1. Opens phone
2. Installs TITAN SOLO PWA (like app, no app store)
3. Logs in
4. Sees "1 job today"
5. Click job → Opens GROUND ZERO dispatch (shows job)
6. Click "Start" → Opens TITAN GO chatbot
7. Chatbot: "Ready to clean?"
8. "Yes" → Shows checklist as conversation
9. Takes photos via chatbot
10. "Done?" → Creates invoice
11. Invoice sent via ZEROPAY (zero fees)
12. Gets paid
13. All on phone, conversational, offline-capable

Apps used: TITAN SOLO (primary) + TITAN GO (chatbot job) + ZEROPAY (payment)
Technical: All PWA, all work offline, smooth experience

GROWING BUSINESS (Day 1)

Scheduler/Dispatch Person:

1. Opens tablet/desktop
2. Opens GROUND ZERO (full-screen SPA)

3. Sees calendar with jobs (drag-drop interface)
4. Click "Unassigned Jobs" card → AI suggests assignments
5. Click "Route Optimization" card → AI reorders route
6. Dispatches jobs to technicians (pushes to their TITAN GO)
7. Real-time updates from field (TITAN GO → GROUND ZERO)
8. Click "Issues" card → See problems, reassign if needed
9. All day on GROUND ZERO, real-time control

Technical: SPA, real-time WebSocket updates, multi-user sync

Field Technician:

1. Gets push notification (job dispatched)
2. Opens TITAN GO PWA on phone
3. Chatbot: "You have job at 123 Main St"
4. Clicks "Navigate" → GPS opens
5. At job, clicks "Start" → Checklist as conversation
6. Chatbot guides every step
7. Takes photos (AI checks quality)
8. Signs off
9. Job marked complete in GROUND ZERO (real-time)

Technical: PWA, offline capability, real-time sync back to dispatch

Customer (Property Manager):

1. Receives SMS with job tracking link
2. Clicks link → ZERO FUSS chatbot opens
3. Chatbot: "Your property is being cleaned"
4. "Where are they?" → Shows map with real-time location
5. Job complete → Photos appear in chat
6. Invoice appears → Payment link
7. Clicks to pay (via ZEROPAY - zero fees)
8. Receipt + tax document auto-generated
9. Can reschedule next cleaning via chat

Technical: PWA or web, chatbot interface, zero forms, conversational

ENTERPRISE OPERATION (Day 1)

Owner/Admin:

1. Opens TITAN PRO dashboard (full-screen SPA)

2. Sees all KPIs across all verticals
3. Click "Residential Revenue" card → AI trend analysis
4. Click "Dispatch Health" card → AI identifies delays
5. Click "Team Utilization" card → AI warns of overload
6. Click "Customer Satisfaction" card → AI lists unhappy customers
7. Click "Overdue Invoices" card → AI suggests collection tactics
8. Click "Compliance" card → AI verifies all jobs adherent
9. All AI-powered, all real-time, all from one dashboard

Technical: React SPA, real-time data, D3 visualizations, AI cards

Lead Gen Specialist:

1. Opens TITAN HELLO (full-screen SPA)
2. Configures campaign (vertical: bond cleaning)
3. Sets targets (real estate agents in [region])
4. Writes outreach template (or uses AI draft)
5. Click "Launch Campaign" → AI scrapes, sends, follows up
6. Click "Campaign Performance" card → AI analyzes conversions
7. Adjusts targeting based on AI insights
8. Books convert automatically into GROUND ZERO

Technical: SPA, web scraping backend, campaign management, real-time tracking

Marketing Agent:

1. Opens TITAN STUDIO (full-screen SPA)
2. Click "Create Social Post" → AI suggests 3 options
3. Picks best, edits text/images
4. Schedules across all platforms
5. Click "Build Landing Page" → AI auto-populates from company data
6. Adds before/after photos
7. Click "Run Campaign" → Sets emails, timing, targeting
8. AI writes content, schedules sends, tracks opens
9. Click "Performance" card → AI recommends improvements

Technical: SPA, template engine, media editor, scheduling, analytics

IV. TECHNICAL ADVANTAGES

PWA vs Native Apps

PWA Advantages (TITAN GO, ZERO FUSS, TITAN ZERO, etc):

- └ No app store approval needed
- └ Instant updates (no approval delays)
- └ Offline-first (works without connection)
- └ Works on all devices (phone, tablet, desktop)
- └ No storage requirements (small footprint)
- └ Same code across platforms
- └ Installable like app (home screen icon)

Traditional App Problems We Avoid:

- └ App store submissions (5-10 days delay)
- └ Version fragmentation (different user versions)
- └ Update delays (some users stuck on old versions)
- └ Platform-specific bugs (iOS vs Android)
- └ App download friction (store, permissions, storage)
- └ Deployment cost (maintaining 2 codebases)

SPA Control Panels vs Traditional Web

SPA Advantages (HELLO, GROUND ZERO, PRO, STUDIO):

- └ Real-time updates (WebSocket connections)
- └ No page reloads (smooth user experience)
- └ Instant responsiveness (AI card interactions)
- └ Desktop/tablet optimized (not mobile)
- └ Multi-user sync (everyone sees updates instantly)
- └ Keyboard shortcuts (power user efficiency)
- └ Data persistence (state maintained in browser)

Offline-First Architecture

Field Staff Problem Solved:

- └ TITAN GO works with no internet (field cleanup sites)
- └ Checklists cached locally
- └ Photos stored locally
- └ Syncs automatically when online
- └ No jobs lost due to connectivity

Traditional App Problem:

- └ Jobber crashes if no connection
 - └ Photos get stuck
 - └ Checklists not available
 - └ Staff frustrated
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V. PRICING WITH CORRECT ARCHITECTURE

Understanding the Apps

PWA APPS (Install on phone, work offline):

- └ TITAN GO (field chatbot)
- └ ZERO FUSS (customer portal)
- └ TITAN ZERO (AI training)
- └ ZEROPAY (payments)
- └ TITAN SOLO (solo tech)
- └ User count: Pay per user (or included in subscriptions)

SPA CONTROL PANELS (Desktop/tablet, real-time):

- └ TITAN HELLO (lead gen management)
- └ GROUND ZERO (dispatch control)
- └ TITAN PRO (admin dashboard)
- └ TITAN STUDIO (marketing automation)
- └ User count: Pay per seat or concurrent users

Pricing Model

SOLO CLEANER:

- └ TITAN SOLO PWA: \$29/month (includes all features)
- └ ZEROPAY: FREE (included)
- └ Total: \$29/month (can use GROUND ZERO if needed)

GROWING BUSINESS:

- └ TITAN HELLO SPA: \$299/month
- └ GROUND ZERO SPA: \$199/month (unlimited dispatch staff)
- └ TITAN GO PWA: \$99/month (one team member, scales with users)
- └ ZERO FUSS PWA: \$99/month (customer-facing)
- └ TITAN ZERO PWA: \$149/month (shared knowledge)
- └ TITAN PRO SPA: \$199/month (admin dashboard)
- └ ZEROPAY PWA: FREE (+ optional PRO \$49)
- └ TITAN STUDIO SPA: \$249/month (one marketing agent)
- └ Total: \$1,193/month for full system

ENTERPRISE:

- └ Add more GROUND ZERO seats (unlimited dispatch staff per seat)
- └ Add more TITAN HELLO campaigns
- └ Add more TITAN STUDIO agents
- └ Add user seats for TITAN GO (scale with team)
- └ TITAN PRO scales (one dashboard, unlimited viewers)
- └ Total: \$1,500-3,000/month for multi-vertical operation

VI. DEPLOYMENT FLOW

New Customer Signup

Day 1: Customer signs up

- | Chooses vertical (residential)
- | Gets GROUND ZERO SPA access (admin)
- | Gets TITAN HELLO SPA (if subscribing)
- | Receives email: "Here's how to use..."

Day 2: Adds first staff member

- | Sends invite link for TITAN GO PWA
- | Staff installs PWA on phone
- | First job dispatched from GROUND ZERO
- | Staff gets notification, opens TITAN GO
- | Job completion via chatbot

Day 3: First customer uses ZERO FUSS

- | Customer gets SMS with tracking link
- | Opens ZERO FUSS chatbot PWA
- | Tracks job, sees photos, pays via ZEROPAY
- | All zero-friction

Week 2: Adds marketing

- | Adds TITAN STUDIO agent
- | Creates social campaigns (AI-assisted)
- | Builds landing page (AI-auto-populated)
- | Tracks leads from TITAN HELLO back to GROUND ZERO
- | Closed loop: marketing → booking → execution → payment

VII. THE COMPLETE TECHNICAL PICTURE

Architecture

FRONTEND LAYER:

- | PWA Apps (React Native Web)
 - | | TITAN GO (field chatbot)
 - | | ZERO FUSS (customer portal)
 - | | TITAN ZERO (AI training)
 - | | ZEROPAY (payments)
 - | | TITAN SOLO (solo tech)

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- └ SPA Control Panels (React SPA)
 - └ TITAN HELLO (lead gen)
 - └ GROUND ZERO (dispatch)
 - └ TITAN PRO (admin dashboard)
 - └ TITAN STUDIO (marketing)

BACKEND LAYER:

- └ API Server (Laravel MagicAI v10)
 - └ Authentication (JWT across all apps)
 - └ Real-time (WebSocket for live updates)
 - └ Offline sync (background sync for PWAs)
 - └ AI Integration (Vertex, Claude, etc)
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- └ Data Layer
 - └ MySQL (persistent data)
 - └ SQLite (offline caching in PWAs)
 - └ Redis (real-time state)
 - └ S3 (photo/document storage)

INFRASTRUCTURE:

- └ Hosting (AWS/GCP)
- └ CDN (CloudFlare)
- └ Service Workers (offline capability)
- └ Push Notifications (job dispatch)

Data Flow

LEAD GEN FLOW:

TITAN HELLO (SPA)

- AI scrapes web
- Sends outreach
- GROUND ZERO (SPA) receives booking
- GROUND ZERO dispatches job
- TITAN GO (PWA) receives notification
- Tech completes job
- ZEROPAY (PWA) collects payment
- ZERO FUSS (PWA) shows customer status
- TITAN PRO (SPA) shows revenue

REAL-TIME DISPATCH FLOW:

GROUND ZERO (SPA)

- ← Job queued
- Tech notification (push to TITAN GO)
- ← TITAN GO confirms received
- GROUND ZERO shows status

- ← TITAN GO job progress
 - Customer notified (ZERO FUSS)
 - ← Job marked complete
 - GROUND ZERO shows completion
 - ZEROPAY auto-sends invoice
 - ← Payment received
 - All KPIs update (TITAN PRO)
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VIII. FINAL SUMMARY

The Product (Corrected)

5 PWA Apps (Phone-based, offline-capable, chatbot interfaces):

- └ TITAN GO (field staff job chatbot)
- └ ZERO FUSS (customer portal chatbot)
- └ TITAN ZERO (vertical AI training chatbot)
- └ ZEROPAY (payment PWA + AI follow-up)
- └ TITAN SOLO (solo tech simplified PWA)

4 SPA Control Panels (Desktop/tablet, real-time, AI-powered cards):

- └ TITAN HELLO (lead gen management)
- └ GROUND ZERO (dispatch control + scheduling)
- └ TITAN PRO (admin dashboard + analytics)
- └ TITAN STUDIO (marketing automation)

All connected:

- └ Real-time sync across all apps
- └ Offline-first for field staff
- └ AI card interactions on control panels
- └ Chatbot interface on customer-facing PWAs
- └ Zero transaction fees on ZEROPAY
- └ Vertical specialization on all apps

Why This Architecture Wins

Better than Jobber:

- └ Offline-capable PWAs (field staff never stuck)
- └ Real-time SPA dispatch (vs refresh-based web)
- └ Chatbot interfaces (vs forms)
- └ AI card interactions (vs dumb data entry)
- └ No app store required (vs app updates taking days)
- └ Better UX (conversational vs transactional)

- └ Mobile-first design (vs desktop-first)

Better than native apps:

- └ No app store approval delays
- └ No platform fragmentation
- └ No version conflicts
- └ No update friction
- └ Same code everywhere
- └ Instant deployment

This is the complete, correct Titan BOS architecture. Ready to build. 🚀